EXP 18: Implement Perceptron based IRIS classification

CODING:

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# Perceptron-Based Iris Classification

# Training data: [Sepal Length, Sepal Width]
X = [
   [5.1, 3.5],
   [4.9, 3.0],
   [5.0, 3.4],
   [6.4, 3.2],
   [6.9, 3.1],
   [5.5, 2.3]
]

# Setosa = 1, Non-Setosa = 0
Y = [1, 1, 1, 0, 0, 0]

# Initialize weights and bias
w1 = 0
w2 = 0
bias = 0
learning_rate = 0.1

# Training the perceptron
for epoch in range(20):
   for i in range(len(X)):
      z = w1 * X[i][0] + w2 * X[i][1] + bias
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        prediction = 1 if z >= 0 else 0

        error = Y[i] - prediction

        w1 += learning_rate * error * X[i][0]
        w2 += learning_rate * error * X[i][1]
        bias += learning_rate * error

# Test a new flower
test = [5.0, 3.4]

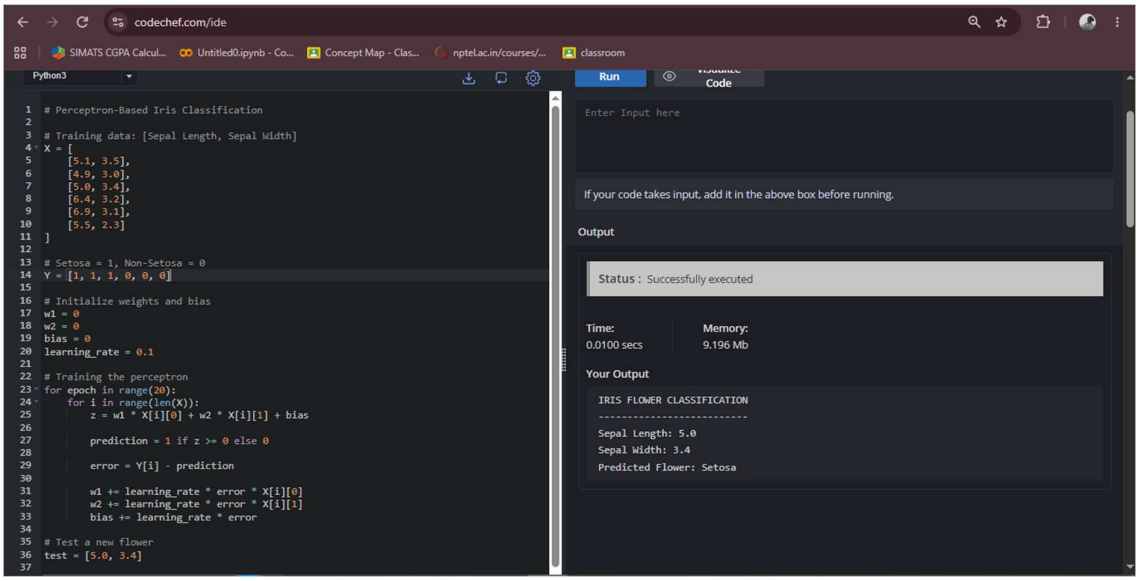
z = w1 * test[0] + w2 * test[1] + bias

result = 1 if z >= 0 else 0

print("IRIS FLOWER CLASSIFICATION")
print("--------------------------")
print("Sepal Length:", test[0])
print("Sepal Width:", test[1])

if result == 1:
   print("Predicted Flower: Setosa")
else:
   print("Predicted Flower: Non-Setosa")
```

OUTPUT:


codechef.com/ide
SIMATS CGPA Calcul...
Untitled0.ipynb - Co...
Concept Map - Clas...
nptel.ac.in/courses/...
classroom
Python3
Run
Code
Perceptron-Based Iris Classification
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X = [
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Setosa = 1, Non-Setosa = 0
Y = [1, 1, 1, 0, 0, 0]
Initialize weights and bias
w1 = 0
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Training the perceptron
for epoch in range(20):
for i in range(len(X)):
z = w1 * X[i][0] + w2 * X[i][1] + bias
prediction = 1 if z >= 0 else 0
error = Y[i] - prediction
w1 += learning_rate * error * X[i][0]
w2 += learning_rate * error * X[i][1]
bias += learning_rate * error
Test a new flower
test = [5.0, 3.4]
Enter Input here
If your code takes input, add it in the above box before running.
Output
Status : Successfully executed
Time:
0.0100 secs
Memory:
9.196 Mb
Your Output
IRIS FLOWER CLASSIFICATION
Sepal Length: 5.0
Sepal Width: 3.4
Predicted Flower: Setosa